

Modeling the Invisible Workshop Report

Adaptive Ensemble Forecasting for Sequential Influenza Hospitalization Prediction

Team-01

1 Workshop and Challenge Overview

The *Modeling the Invisible Workshop* was organized as a real-time forecasting competition focused on predicting influenza-associated hospitalization dynamics from limited sequential data. The general goal was to simulate a realistic public-health forecasting situation in which only partial time-series data are available at each decision point. Teams received updated releases of hospitalization data and then submitted forecasts for the next several weeks. The workshop emphasized scientific modeling, reproducible computational workflows, uncertainty-aware prediction, and collaboration under data scarcity.

The key forecast target was the weekly hospitalization rate per 100,000 people. In each round, the organizers released a prefix of the epidemic trajectory, and teams predicted future hospitalization values. The submission format included the week number and predicted hospitalizations per 100,000; an optional third column contained an estimate of the effective reproduction number, R_t . Forecast accuracy was evaluated using root mean square error (RMSE), computed between the submitted predictions and the subsequently released ground-truth values:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n \left(\hat{H}_i - H_i \right)^2},$$

where $\hat{H}_i$ is the forecasted hospitalization rate and H_i is the realized value. In the workshop setting, lower RMSE indicates better forecasting performance. The workshop website also emphasized that the actual challenge data could contain more than one peak, and that noisy data would be included. Therefore, a successful model needed to be adaptive rather than fixed to a single epidemic shape.

2 Model Structure

Our final modeling strategy was based on an adaptive ensemble. Instead of relying on one forecasting assumption, the model generated several candidate forecasts and combined them using data-dependent weights. Let

$$H_{1:t} = (H_1, H_2, \dots, H_t)$$

be the released hospitalization trajectory at a given round, and let K be the forecast horizon. Component j produces a forecast

$$\hat{H}_{t+1:t+K}^{(j)} = \left(\hat{H}_{t+1}^{(j)}, \dots, \hat{H}_{t+K}^{(j)} \right).$$

The final ensemble forecast is

$$\hat{H}_{t+1:t+K} = \sum_{j=1}^m w_j \hat{H}_{t+1:t+K}^{(j)},$$

where $w_j \geq 0$ and

$$\sum_{j=1}^m w_j = 1.$$

The central advantage of this ensemble structure is robustness across regimes. A pure log-linear model may work well during early growth but can overpredict near a peak. A last-value model is stable during flat periods but fails during fast growth. A peak-turn model protects against overshooting but may become too conservative during rebound phases. The ensemble avoids committing completely to one assumption.

2.1 Forecasting Components

The model used the following main components.

Last-value baseline. This component carries the most recent observation forward:

$$\hat{H}_{t+k}^{(\text{last})} = H_t.$$

It is simple, stable, and useful when the released trajectory is short or noisy.

Damped-growth component. This component estimates recent log-growth but gradually weakens that growth over the forecast horizon. It captures increasing or decreasing trends while avoiding unlimited exponential extrapolation:

$$\hat{H}_{t+k}^{(\text{damped})} = \hat{H}_{t+k-1}^{(\text{damped})} \exp(g_k),$$

where g_k is a damped version of recent growth. This component is useful for smooth growth and smooth decline.

Log-linear component. This component fits a line to recent log-hospitalization values:

$$\log(H_i + \epsilon) \approx a + bi.$$

It is useful when the epidemic is in an approximately exponential growth or decay phase. However, because it can overpredict near peaks, it is controlled through clipping and adaptive weighting.

Log-quadratic peak-turn component. This component fits a quadratic curve on the log scale:

$$\log(H_i + \epsilon) \approx a + bi + ci^2.$$

The quadratic term helps detect curvature. When growth begins to decelerate, this component can represent an approaching peak and transition into decline. It was especially important for preventing the model from continuing explosive growth after the curve had already reached a high value.

Square-root cubic component. This component models the transformed series

$$S_i = \sqrt{H_i}$$

with a local cubic polynomial and transforms the forecast back by squaring. The square-root scale reduces the influence of very large hospitalization values and provides flexible nonlinear curve-fitting.

Analog-template component. This component compares the current partial curve with previously seen epidemic shapes. If the current trajectory resembles a segment of a previous curve, the future of that previous curve is used as a template. This component is useful when the epidemic follows a realistic repeated pattern, such as a single peak, long decline, or multi-wave structure.

SIR mechanistic component. The ensemble also included a mechanistic SIR component. The SIR model divides the population into susceptible, infectious, and recovered compartments:

$$\frac{dS}{dt} = -\frac{\beta SI}{N}, \quad \frac{dI}{dt} = \frac{\beta SI}{N} - \gamma I, \quad \frac{dR}{dt} = \gamma I.$$

The SIR component fits transmission parameters to the observed hospitalization trajectory and then forecasts future hospitalizations. It is useful because it provides a biologically constrained forecast: growth cannot continue indefinitely because the susceptible pool is depleted. It also gives a principled estimate of the effective reproduction number,

$$R_t = \frac{\beta S(t)}{\gamma N}.$$

Values $R_t > 1$ indicate growth, while $R_t < 1$ indicate decline.

2.2 Adaptive Weighting

Weights were assigned using two ideas. First, the model performed internal rolling validation using only the released data. For example, if data were available through week t , the model temporarily hid the last few released points, predicted them using earlier points, and measured the component errors. Components with lower internal RMSE received larger preliminary weights:

$$\tilde{w}_j \propto \frac{1}{(S_j + \eta)^2},$$

where S_j is a robust internal validation score for component j , and η prevents extreme weights.

Second, the preliminary weights were adjusted using regime detection. If the curve showed active growth, the model increased the weights of growth-sensitive components. If the curve was near a peak, the model boosted peak-turn and SIR components. If the curve was declining, the model emphasized damped-growth and post-peak components. If the data were noisy, reactive components were downweighted and robust trend components were favored. After these adjustments, weights were normalized so they summed to one.

During the workshop we used several model versions because the challenges changed qualitatively. Challenge 1 mainly required adaptation to smooth growth, delayed acceleration, and post-peak decline. Challenge 2 required additional handling of noisy data, abrupt latent surge, sudden peak collapse, rebound, and possible late second-wave behavior. Although different versions were used, all versions followed the same core ensemble principle.

3 Results

3.1 Challenge 1

Challenge 1 showed a relatively smooth epidemic trajectory with gradual early growth, followed by a stronger growth phase, a peak, and then decline. In the first round, the model performed well because the observed data already showed a coherent increasing trend and the forecast extended it

smoothly. In the second round, the model became too conservative and underestimated the later acceleration. In the third round, after seeing rapid growth, the model overestimated continued growth and missed the sharp turning point. In the fourth round, the model correctly predicted decline but still tended to overestimate the tail, indicating that the model had learned the direction but not the full speed of decay.

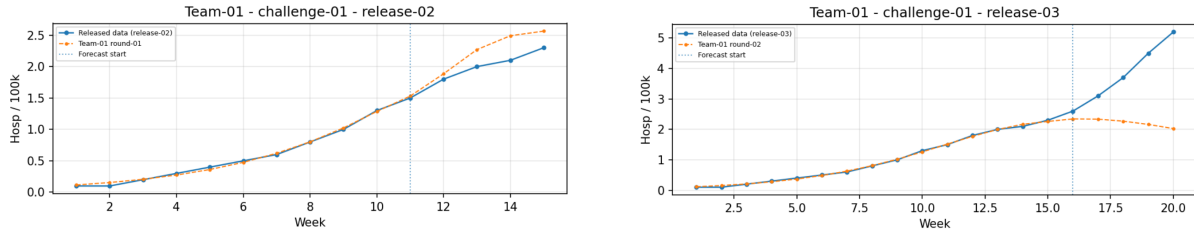


Figure 1: Challenge 1, rounds 1 and 2. In round 1, the forecast followed the released trajectory well and slightly overpredicted the continued increase. In round 2, the model underestimated the later acceleration, showing the difficulty of detecting a transition from moderate growth to stronger growth from limited data.

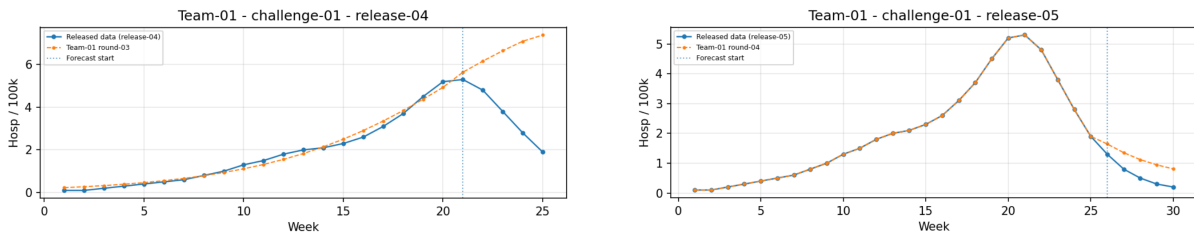


Figure 2: Challenge 1, rounds 3 and 4. In round 3, the model extrapolated the recent growth too strongly and missed the peak-turn. In round 4, after the decline became visible, the model predicted the correct downward direction but overestimated the declining tail.

3.2 Challenge 2

Challenge 2 was more difficult because the data were noisy and contained a dramatic delayed surge after an initially flat period. In the first round, a conservative model would have predicted nearly flat values, but we used a latent-surge prior because the challenge allowed multi-agent or delayed epidemic behavior. This helped capture the sudden increase. In the second round, the model overestimated persistence near the peak; the realized data declined sharply after the surge. In the third round, the model adapted to the post-surge rebound-and-decay pattern and closely followed the realized trajectory. In the fourth round, the model used a guarded late-second-wave component, but the realized data declined faster than the forecast. Overall, Challenge 2 emphasized that noisy data and multi-wave dynamics require both flexibility and caution.

4 Discussion

The workshop demonstrated that no single model is reliable across all forecasting regimes. Trend extrapolation is useful during smooth growth but dangerous near peaks. Peak-turn models are safer after rapid growth but may overreact if a second wave occurs. Mechanistic SIR modeling

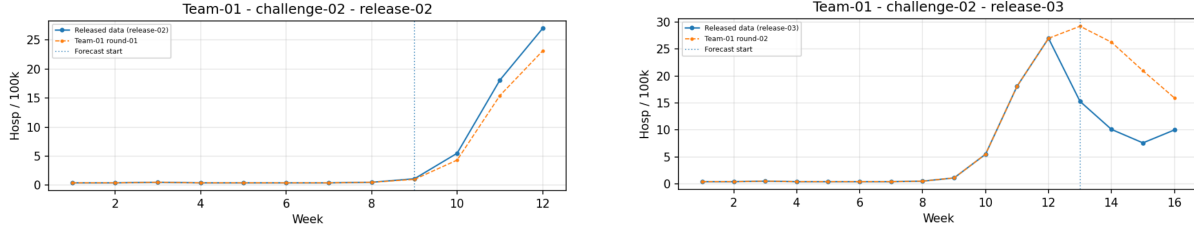


Figure 3: Challenge 2, rounds 1 and 2. Round 1 shows the latent-surge scenario: the early trajectory was nearly flat, but the true trajectory rose sharply. The aggressive latent-surge prior captured the timing of the surge, though it underpredicted the final magnitude. In round 2, the model expected a short plateau or gradual decline, but the released data showed a much faster post-peak drop.

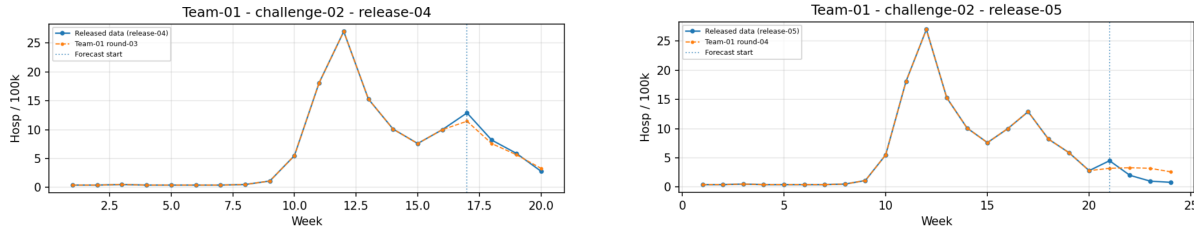


Figure 4: Challenge 2, rounds 3 and 4. In round 3, the model captured the rebound-and-decay pattern well after the sudden drop. In round 4, the model allowed a small late second-wave or plateau, but the true trajectory declined faster, producing an overprediction of the tail.

provides useful biological constraints, but fitting it to sparse or noisy hospitalization data can still be difficult. The adaptive ensemble approach was therefore appropriate because it allowed different components to dominate in different regimes.

Our main successes were the ability to adapt the model across releases, incorporate mechanistic SIR reasoning, and create specialized components for latent surge, post-surge decay, and possible late rebound. The main failure modes were delayed recognition of abrupt regime changes: underpredicting acceleration in Challenge 1, overpredicting after a peak in Challenge 2, and sometimes overestimating declining tails. These results suggest that future versions should include better probabilistic regime-switching, explicit noise models, and calibrated uncertainty intervals rather than only component-disagreement intervals.

5 Conclusion

We developed an adaptive ensemble forecasting model for sequential influenza hospitalization prediction. The model combined simple baselines, statistical trend models, peak-turn detection, analog templates, and a mechanistic SIR component. Weights were assigned using internal validation and regime-aware adjustment. Across the two challenges, the model performed best when the current regime was correctly identified and worst when the epidemic changed direction abruptly. The final workflow provided a flexible and interpretable framework for real-time epidemic forecasting under limited and noisy data.